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Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Salary Distribution
- Features Correlated to Salary
- Regression Results
- Interpretation and Next Steps

Data Description

The dataset contains **373 employees** across 6 variables: Age, Gender, Education Level, Job Title, Years of Experience, and Salary.

The original dataset had **375 rows**. A missing-value check revealed **12 missing values** (2 per column across all 6 columns). All rows containing at least one missing value were removed using **listwise deletion**, leaving **373 complete records** for analysis.

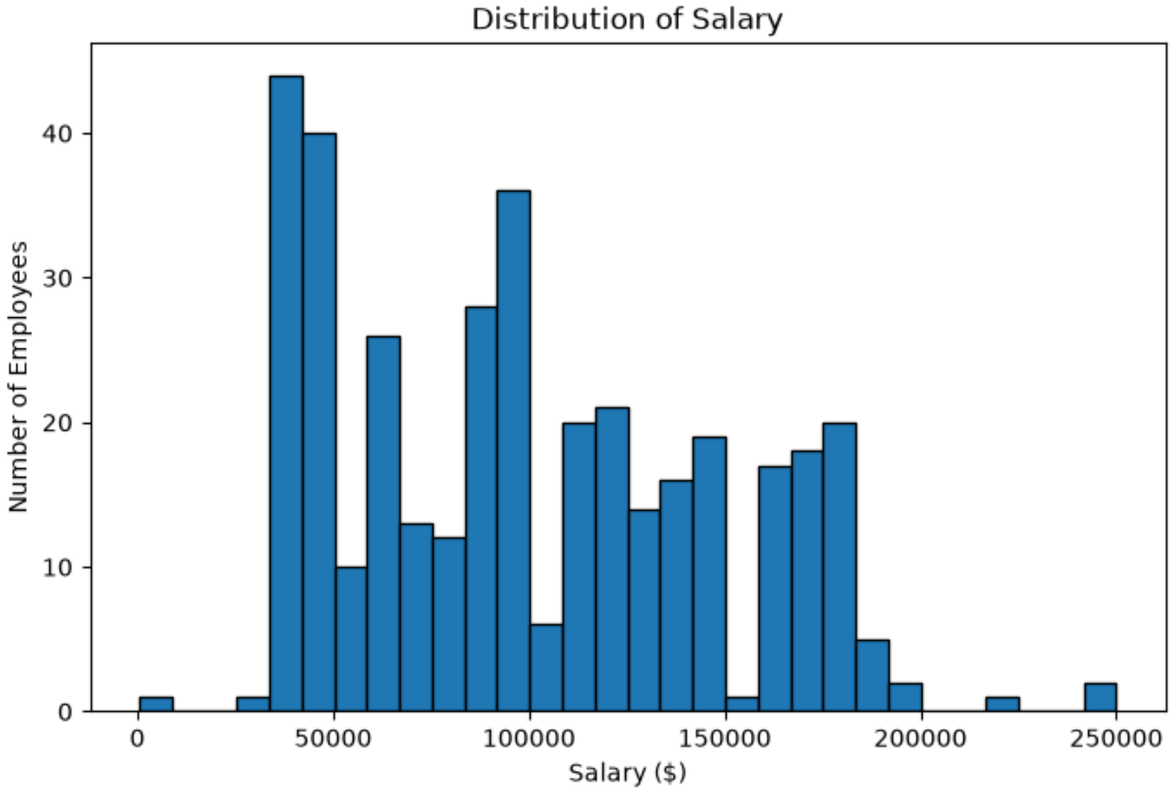
Education breakdown:

- Bachelor's: **224 employees**
- Master's: **98 employees**
- PhD: **51 employees**

Salary Distribution

Statistic	Value
Minimum	$350
25th Percentile	$55,000
Median	$95,000
Mean	$100,577
75th Percentile	$140,000
Maximum	$250,000
Std. Deviation	$48,240

The distribution is **right-skewed**: most employees earn between $55,000 and $140,000, but a long upper tail pulls the mean above the median.



Features Correlated to Salary

Years of Experience shows the strongest linea relationship with salary (Pearson r = **0.93**).

Education Level shows a clear salary gradient:

- PhD average: **$157,843**
- Master's average: **$129,796**
- Bachelor's average: **$74,756**

Gender: Male employees averaged **$103,868** vs. Female at **$97,011** - a gap of ~$6,857.

Job title keywords (Director, Senior, Manager) are also associated with above-average salaries as confirmed by the regression results on the following slides.



Regression Results - Full Model

A linear regression model was fit using **10 predictors**: is_male, is_director, is_junior, is_senior, is_manager, is_analyst, is_engineer, Years of Experience, is_masters, is_phd.

R^2 = 0.915 | Adjusted R^2 = 0.913

Three variables were **not statistically significant** at the 5% level and were removed:

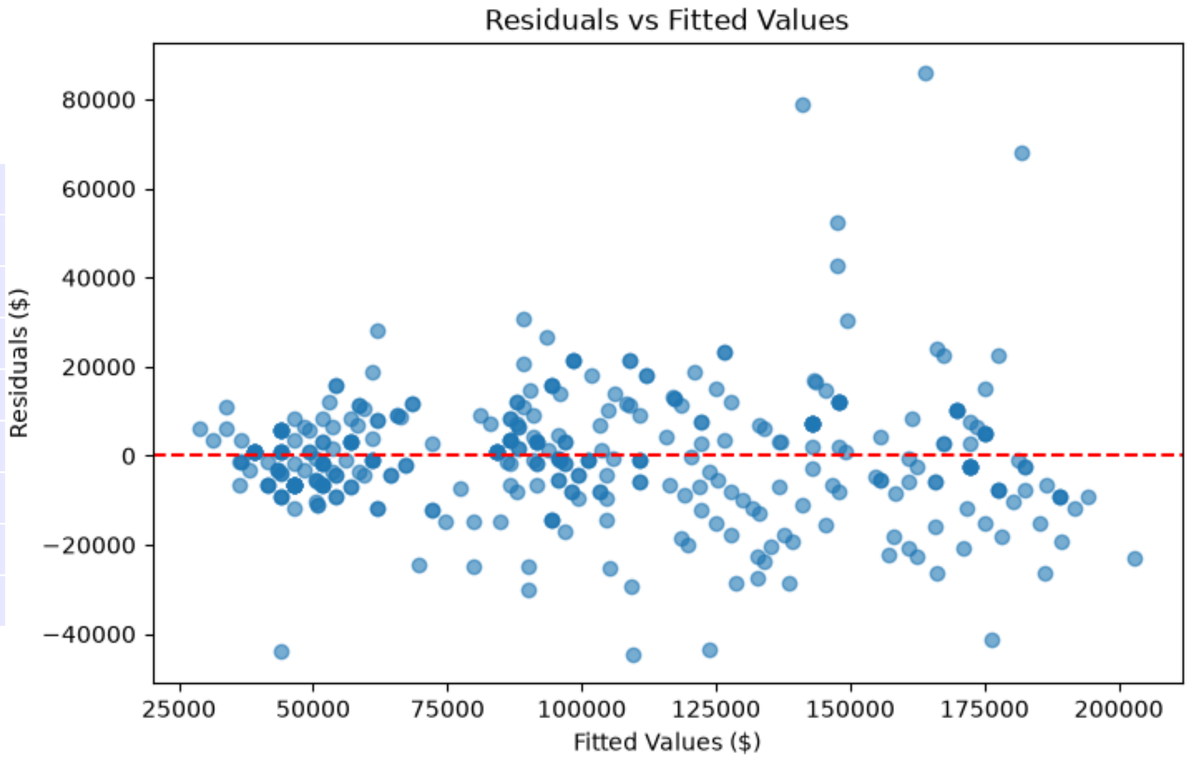
- is_junior - p = 0.068
- is_analyst - p = 0.933
- is_engineer - p = 0.508

Regression Results - Reduced Model

After removing the three insignificant predictors, the final model contains **7 predictors**. All are statistically significant ($p < 0.05$).

Variable	Coefficient	p-value	95% Confidence Interval
Intercept	$28,760	< 0.001	[$25,460 , $32,060]
is_male	$7,630	< 0.001	[$4,716 , $10,544]
is_director	$24,980	< 0.001	[$18,100 , $31,860]
is_senior	$14,450	< 0.001	[$10,870 , $18,020]
is_manager	$4,285	0.019	[$701 , $7,869]
Years of Experience	$5,104	< 0.001	[$4,756 , $5,453]
is_masters	$14,040	< 0.001	[$10,045 , $18,035]
is_phd	$23,081	< 0.001	[$17,652 , $28,510]

Model Fit: $R^2 = 0.914$ | Adjusted $R^2 = 0.913$
The model explains approximately **91.4% of the variation in salary**, indicating a strong overall fit.



Interpretation - Years of Experience

- **Expected change in salary:**
  Each additional year of experience is associated with an expected salary increase of **$5,104**, holding gender, job title, and education level constant.
- **Confidence in this estimate:**
  We can be **95% confident** that the true per-year salary increase is between **$4,756 and $5,453**.
- Because this interval lies entirely above zero and the p-value is < 0.001, the relationship is highly statistically significant. In practical terms, a 5-year difference in experience translates to an expected salary difference of roughly **$25,500**.

Interpretation - Education Level (PhD)

- **Expected change in salary:**
  Holding all other variables constant, an employee with a **PhD** is expected to earn **$23,081 more** than an employee whose highest degree is a **Bachelor's** (the baseline group).
- **Confidence in this estimate:**
  We can be **95% confident** that the true PhD salary premium over a Bachelor's degree falls between **$17,652 and $28,510**.
- The result is statistically significant ($p < 0.001$). In practical terms, a PhD premium represents approximately **23% of the dataset's average salary** of $100,577 - a substantial economic return to doctoral education in this sample.